Core CGE and GTAP-E Equations

Mathematical Reference for RunGTAP Simulations

1 General CGE Representation

A computable general equilibrium (CGE) model can be represented compactly as

$$\boxed{F\left(\mathbf{x}^{\text{ENDO}}, \mathbf{z}^{\text{EXO}}; \boldsymbol{\theta}\right) = 0} \tag{1}$$

where

$$\begin{aligned} \mathbf{x}^{\text{ENDO}} &= \text{vector of endogenous variables,} \\ \mathbf{z}^{\text{EXO}} &= \text{vector of exogenous variables,} \\ \boldsymbol{\theta} &= \text{vector of behavioral parameters.} \end{aligned}$$

For GTAP-E, the parameter vector may include

$$\boldsymbol{\theta} = \{ESUBD,\ ESUBM,\ ESUBVA,\ ETRAE,\ ELPENY,\ ELGENY,\ ELFENY,\ ELFKEN,\ SUBPAR,\ INC \tag{2}$$

2 Production

Let sector j in region r produce output using primary factors, energy, and intermediate inputs:

$$\boxed{Q_{jr} = F_{jr}\left(K_{jr}, L_{jr}, N_{jr}, E_{jr}, X_{1jr}, \ldots, X_{Ijr}; A_{jr}\right)} \tag{3}$$

where A_{jr} denotes technology.
Under constant returns to scale (CRS),

$$\boxed{F_{jr}(\lambda \mathbf{X}) = \lambda F_{jr}(\mathbf{X}), \qquad \lambda > 0.} \tag{4}$$

3 Zero-Profit Condition

Under perfect competition and constant returns to scale,

$$\boxed{P_{jr} Q_{jr} = \sum_i P_{ijr} X_{ijr} + \sum_f W_{fjr} F_{fjr}.} \tag{5}$$

Equivalently, output price equals unit cost:

$$\boxed{P_{jr} = C_{jr}\left(\mathbf{P}_r, \mathbf{W}_r; A_{jr}\right).} \tag{6}$$

4 CES Value-Added Production

A generic CES value-added composite can be written as

$$VA_{jr} = A_{jr}\left[\sum_{f}\delta_{fjr}F_{fjr}^{\frac{\sigma_{VA,j}-1}{\sigma_{VA,j}}}\right]^{\frac{\sigma_{VA,j}}{\sigma_{VA,j}-1}}. \tag{7}$$

The elasticity of substitution is

$$\boxed{\sigma_{VA,j} = ESUBVA_j.} \tag{8}$$

A corresponding percentage-change factor-demand relationship is

$$\boxed{q_{fjr} = q_{VA,jr} - \sigma_{VA,j}\left(p_{fjr} - p_{VA,jr}\right).} \tag{9}$$

5 GTAP-E Energy Nest

The energy composite can be represented as

$$E_{jr} = \text{CES}\left(E_{1jr}, E_{2jr}, \ldots, E_{Njr}; \sigma_{jr}^{E}\right), \tag{10}$$

where

$$\boxed{\sigma_{jr}^{E} = ELFENY(j,r).} \tag{11}$$

The GTAP-E percentage-change demand equation is

$$\boxed{qf(i,j,r) = -af(i,j,r) + qf(\text{eny},j,r) - ELFENY(j,r)\left[pf(i,j,r) - af(i,j,r) - pf(\text{eny},j,r)\right].} \tag{12}$$

The price of the energy composite is

$$\boxed{pf(\text{eny},j,r) = \sum_{k\in ENY_FCOMM} FSHENY(k,j,r)\left[pf(k,j,r) - af(k,j,r)\right].} \tag{13}$$

6 Capital–Energy Nest

Capital and energy are combined in a CES subproduction nest:

$$\boxed{KE_{jr} = \text{CES}\left(K_{jr}, E_{jr}; \sigma_{jr}^{KE}\right).} \tag{14}$$

where

$$\boxed{\sigma_{jr}^{KE} = ELFKEN(j,r).} \tag{15}$$

The corresponding demand relationship is

$$\boxed{qf(i,j,r) = -af(i,j,r) + qf(\text{ken},j,r) - ELFKEN(j,r)\left[pf(i,j,r) - af(i,j,r) - pf(\text{ken},j,r)\right],} \tag{16}$$

for $i \in KEN_FCOMM$.

7 Armington Structure

GTAP assumes that commodities are differentiated by origin.

7.1 Domestic Goods versus Aggregate Imports

The first Armington nest is

$$\boxed{Q_{ir} = \left[\delta_{ir}^{D}(Q_{ir}^{D})^{\rho_i} + \delta_{ir}^{M}(Q_{ir}^{M})^{\rho_i}\right]^{1/\rho_i},} \tag{17}$$

where

$$\rho_i = \frac{ESUBD_i - 1}{ESUBD_i}. \tag{18}$$

Thus,

$$\boxed{ESUBD_i = \text{elasticity of substitution between domestic and imported goods.}} \tag{19}$$

For private consumption, GTAP-E contains

$$\boxed{qpd(i,s) = qpt(i,s) + ESUBD(i)\left[ppt(i,s) - ppd(i,s)\right].} \tag{20}$$

Aggregate imported demand is

$$\boxed{qpm(i,s) = qpt(i,s) + ESUBD(i)\left[ppt(i,s) - ppm(i,s)\right].} \tag{21}$$

The composite price is

$$\boxed{ppt(i,s) = PMSHR(i,s)\,ppm(i,s) + \left[1 - PMSHR(i,s)\right]ppd(i,s).} \tag{22}$$

7.2 Import Source Substitution

Aggregate imports from different source regions are represented as

$$\boxed{Q_{is}^{M} = \left[\sum_{r} \beta_{irs} Q_{irs}^{\frac{\sigma_i^M - 1}{\sigma_i^M}}\right]^{\frac{\sigma_i^M}{\sigma_i^M - 1}}.} \tag{23}$$

where

$$\boxed{\sigma_i^M = ESUBM_i.} \tag{24}$$

Hence,

$$ESUBD_i: \quad Q_{is}^{D} \leftrightarrow Q_{is}^{M}, \tag{25}$$

whereas

$$ESUBM_i: \quad Q_{i,USA,s} \leftrightarrow Q_{i,ROW,s} \leftrightarrow Q_{i,EU,s} \leftrightarrow \cdots. \tag{26}$$

8 Bilateral Trade

Define

$$QXS_{irs} = \text{quantity of commodity } i \text{ exported from } r \text{ to } s. \tag{27}$$

The bilateral export value at world prices is

$$\boxed{VXWD_{irs} = PFOB_{irs} QXS_{irs}.} \tag{28}$$

Imports valued at world/CIF prices satisfy

$$\boxed{VIWS_{irs} = PCIF_{irs} QXS_{irs}.} \tag{29}$$

For example,

$$\boxed{qxs(\text{Oil}, DVCUSA, DVCEU)} \tag{30}$$

represents the percentage change in bilateral oil shipments from the United States to the European Union.

9 Private Household Demand: CDE System

GTAP uses a non-homothetic Constant Difference of Elasticities (CDE) private household demand system.

Define

$$\boxed{\alpha_{ir} = 1 - SUBPAR(i,r).} \tag{31}$$

The Allen partial elasticity of substitution is

$$\boxed{APE(i,k,r) = \alpha_{ir} + \alpha_{kr} - \sum_n PSHUP(n,r)\alpha_{nr}.} \tag{32}$$

For the own-good term,

$$\boxed{APE(i,i,r) = 2\alpha_{ir} - \sum_n PSHUP(n,r)\alpha_{nr} - \frac{\alpha_{ir}}{PSHUP(i,r)}.} \tag{33}$$

Private expenditure's utility elasticity is

$$\boxed{UELASPRIV(r) = \sum_i PSHUP(i,r) INCPAR(i,r).} \tag{34}$$

The income elasticity is

$$EY(i,r) = \frac{INCPAR(i,r)[1-\alpha_{ir}] + \sum_n PSHUP(n,r) INCPAR(n,r)\alpha_{nr}}{\sum_n PSHUP(n,r) INCPAR(n,r)} + \left[\alpha_{ir} - \sum_n PSHUP(n,r)\alpha_{nr}\right]. \tag{35}$$

The uncompensated price elasticity is

$$\boxed{EP(i,k,r) = [APE(i,k,r) - EY(i,r)]\, PSHUP(k,r).} \tag{36}$$

Private household demand is therefore

$$\boxed{qp(i,r) - pop(r) = \sum_k EP(i,k,r)pp(k,r) + EY(i,r)\,[yp(r) - pop(r)]\,.} \tag{37}$$

Notice that the equation is expressed on a **per-capita basis**.

10 Private Household Energy Consumption

The household energy price index is

$$\boxed{pp(\text{eny},r) = \sum_{k \in ENY_PCOMM} PSHENY(k,r)pp(k,r).} \tag{38}$$

Household demand for individual energy commodities is

$$\boxed{qp(i,r) = qp(\text{eny},r) - ELPENY(r)\,[pp(i,r) - pp(\text{eny},r)]\,.} \tag{39}$$

Thus,

$$\boxed{ELPENY(r) = \text{elasticity of substitution within private energy consumption.}} \tag{40}$$

11 Government Energy Demand

The government energy composite price is

$$\boxed{pg(\text{eny},r) = \sum_{k \in ENY_GCOMM} GSHENY(k,r)pg(k,r).} \tag{41}$$

Government energy demand is

$$\boxed{qg(i,r) = qg(\text{eny},r) - ELGENY(r)\,[pg(i,r) - pg(\text{eny},r)]\,.} \tag{42}$$

Thus,

$$\boxed{ELGENY(r) = \text{government energy substitution elasticity.}} \tag{43}$$

12 Private Consumption Utility

The private consumption price index is

$$\boxed{ppriv(r) = \sum_i PSHUP(i,r)pp(i,r).} \tag{44}$$

Private utility satisfies

$$\boxed{yp(r) - pop(r) = ppriv(r) + UELASPRIV(r)\, up(r).} \tag{45}$$

Therefore,

$$\boxed{up(r) = \frac{yp(r) - pop(r) - ppriv(r)}{UELASPRIV(r)}.} \tag{46}$$

This is why population enters the private-utility calculation through per-capita expenditure.

13 Aggregate Regional Household Utility

Aggregate regional household utility is

$$\begin{aligned} u(r) = au(r) &+ DPARPRIV(r) \ln[UTILPRIV(r)]\, dppriv(r) \\ &+ DPARGOV(r) \ln[UTILGOV(r)]\, dpgov(r) \\ &+ DPARSAVE(r) \ln[UTILSAVE(r)]\, dpsave(r) \\ &+ \frac{1}{UTILELAS(r)} \left[y(r) - pop(r) - p(r)\right]. \end{aligned} \tag{47}$$

Hence,

$$\boxed{up(r) \neq u(r) \neq EV(r).} \tag{48}$$

Here,

$$\begin{aligned} up(r) &= \text{private consumption utility change}, \\ u(r) &= \text{aggregate regional household utility change}, \\ EV(r) &= \text{equivalent variation welfare measure}. \end{aligned}$$

14 Mobile versus Sluggish Endowments

In this GTAP-E implementation, the parameter $SLUG(i)$ classifies endowments:

$$\boxed{SLUG(i) = \begin{cases} 0, & \text{mobile endowment}, \\ 1, & \text{sluggish endowment}. \end{cases}} \tag{49}$$

Thus,

$$ENDWS_COMM = \{i \in ENDW_COMM : SLUG(i) > 0\}. \tag{50}$$

The composite price of a sluggish endowment is

$$\boxed{pm(i,r) = \sum_{k \in PROD_COMM} REVSHR(i,k,r) pmes(i,k,r).} \tag{51}$$

The allocation of a sluggish endowment across sectors is

$$\boxed{qoes(i,j,r) = qo(i,r) - endwslack(i,r) + ETRAE(i)\left[pm(i,r) - pmes(i,j,r)\right].} \tag{52}$$

where

$$\boxed{ETRAE(i) \leq 0.} \tag{53}$$

Therefore,

$$\boxed{SLUG = \text{factor classification}, \qquad ETRAE = \text{factor mobility response.}} \tag{54}$$

15 Endowment Prices

Pre-tax and post-tax endowment supply prices are linked by

$$\boxed{ps(i,r) = to(i,r) + pm(i,r).} \tag{55}$$

16 Domestic Commodity Market Clearing

In general,

$$\boxed{Q_{ir}^{Supply} = Q_{ir}^{Demand}.} \tag{56}$$

The GTAP percentage-change domestic-market-clearing condition is

$$\boxed{qds(i,r) = \sum_{j \in PROD_COMM} SHRDFM(i,j,r)qfd(i,j,r) + SHRDPM(i,r)qpd(i,r) + SHRDGM(i,r)qgd(i,r)} \tag{57}$$

or conceptually,

$$\boxed{\text{Domestic Sales} = \text{Firm Demand} + \text{Private Demand} + \text{Government Demand.}} \tag{58}$$

17 Factor Market Clearing

For a mobile primary factor,

$$\boxed{F_{fr}^{S} = \sum_{j} F_{fjr}^{D}.} \tag{59}$$

Thus total regional factor supply must equal factor demand across producing sectors.

18 Capital Accumulation

In levels,

$$\boxed{K_r^E = K_r^B + NetInvestment_r.} \tag{60}$$

The GTAP percentage-change equation is

$$\boxed{ke(r) = INVKERATIO(r)qcgds(r) + [1 - INVKERATIO(r)]\, kb(r).} \tag{61}$$

19 Current and Expected Rates of Return

The current rate of return on capital is

$$\boxed{rorc(r) = GRNETRATIO(r)\,[rental(r) - pcgds(r)]\,.} \tag{62}$$

Expected return is

$$\boxed{rore(r) = rorc(r) - RORFLEX(r)\,[ke(r) - kb(r)]\,.} \tag{63}$$

Hence,

$$\boxed{RORFLEX(r) = \text{flexibility of expected return with respect to capital accumulation.}} \tag{64}$$

20 Baldwin-Type Capital Accumulation Effect

GTAP defines the expansion of investment relative to the capital endowment as

$$\boxed{EXPAND(i,r) = qcgds(r) - qo(i,r), \qquad i \in ENDWC_COMM.} \tag{65}$$

21 Carbon Dioxide Emissions

For private consumption of domestic products,

$$\boxed{gco2pd(i,r) = qpd(i,r).} \tag{66}$$

For private consumption of imported products,

$$\boxed{gco2pm(i,r) = qpm(i,r).} \tag{67}$$

For government consumption,

$$\boxed{gco2gd(i,r) = qgd(i,r)} \tag{68}$$

$$\boxed{gco2gm(i,r) = qgm(i,r)} \tag{69}$$

For firms,

$$\boxed{gco2fd(i,j,r) = qfd(i,j,r)} \tag{70}$$

$$\boxed{gco2fm(i,j,r) = qfm(i,j,r).} \tag{71}$$

22 Commodity-Specific Regional CO_2 Emissions

For $i \in CTAX_COMM$,

$$\begin{aligned} CO2(r,i)gco2(r,i) = \sum_{j \in PROD_COMM} \Big[& CO2IF(i,j,r)gco2fm(i,j,r) \\ & + CO2DF(i,j,r)gco2fd(i,j,r)\Big] \\ & + CO2DG(i,r)gco2gd(i,r) + CO2IG(i,r)gco2gm(i,r) \\ & + CO2DP(i,r)gco2pd(i,r) + CO2IP(i,r)gco2pm(i,r). \end{aligned} \tag{72}$$

23 Total Regional CO_2 Emissions

Total regional emissions satisfy

$$\boxed{CO2T(r)gco2t(r) = \sum_{i \in CTAX_COMM} CO2(r,i)gco2(r,i).} \tag{73}$$

Here $CO2T(r)$ is measured in millions of tonnes of CO_2 in the benchmark database.

24 World CO_2 Emissions

World emissions are

$$\boxed{\left[\sum_r CO2T(r)\right] gco2tw = \sum_r CO2T(r)gco2t(r).} \tag{74}$$

25 Emissions Quotas

Let

$$CO2Q(r) = \text{regional emissions quota.} \tag{75}$$

The power of emissions purchases is determined by

$$\boxed{pemp(r) = gco2t(r) - gco2q(r).} \tag{76}$$

At the trading-bloc level,

$$\boxed{CO2TB(b)gco2tb(b) = \sum_{r:\, REGTOBLOC(r)=b} CO2T(r)gco2t(r).} \tag{77}$$

26 Carbon Tax

The real carbon tax is related to the nominal carbon tax by

$$\boxed{RCTAX(r) = \frac{1}{PIND(r)}\left[NCTAXB(REGTOBLOC(r)) - 0.01\, NCTAXLEV(r)p(r)\right].} \quad (78)$$

The model therefore distinguishes between

$$\begin{aligned} NCTAXLEV(r) &= \text{benchmark nominal carbon-tax level,} \\ NCTAXB(b) &= \text{change in nominal carbon tax,} \\ RCTAX(r) &= \text{change in real carbon-tax rate.} \end{aligned}$$

27 Equivalent Variation

At the conceptual level, equivalent variation can be written as

$$\boxed{EV_r = E(P_r^0, U_r^1) - E(P_r^0, U_r^0),} \quad (79)$$

where

$$\begin{aligned} P_r^0 &= \text{benchmark prices,} \\ U_r^0 &= \text{benchmark utility,} \\ U_r^1 &= \text{counterfactual utility.} \end{aligned}$$

Thus,

$$EV_r > 0 \quad \Rightarrow \quad \text{welfare gain,} \quad (80)$$

and

$$EV_r < 0 \quad \Rightarrow \quad \text{welfare loss.} \quad (81)$$

28 Closure, Shock, and Swap

The complete equilibrium system can be represented as

$$\boxed{F(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}) = 0,} \quad (82)$$

where

$$\begin{aligned} \mathbf{x} &= \text{endogenous variables,} \\ \mathbf{z} &= \text{exogenous variables,} \\ \boldsymbol{\theta} &= \text{parameters.} \end{aligned}$$

A shock changes an exogenous variable:

$$\boxed{\Delta z_k = \bar{z}_k.} \tag{83}$$

For example,

$$\boxed{pop(DVCUSA) = 2} \tag{84}$$

means a 2 percent increase in U.S. population.
A swap changes the closure:

$$\boxed{x_j : ENDO \rightarrow EXO} \tag{85}$$

and simultaneously

$$\boxed{z_k : EXO \rightarrow ENDO.} \tag{86}$$

For a scalar swap,

$$\boxed{\Delta N_{\text{EXO}} = \Delta N_{\text{ENDO}},} \tag{87}$$

so that the degrees of freedom of the model are preserved.

29 Complete CGE System: Compact Summary

The major relationships can be summarized as

$$Q_{jr} = F(K, L, N, E, M; A) \quad \text{Production,} \tag{88}$$

$$P_{jr} = C_{jr}(\mathbf{p}, \mathbf{w}; A) \quad \text{Zero profit,} \tag{89}$$

$$F_{fjr} = F_{fjr}\left(Q_{jr}, \frac{w_{fjr}}{P_{VA,jr}}; \sigma_{VA}\right) \quad \text{Factor demand,} \tag{90}$$

$$E_{jr} = \text{CES}(Electricity, Fuels; \sigma_E) \quad \text{Energy nest,} \tag{91}$$

$$KE_{jr} = \text{CES}(K, E; \sigma_{KE}) \quad \text{Capital–energy nest,} \tag{92}$$

$$Q_{ir} = \text{CES}(Q^D_{ir}, Q^M_{ir}; ESUBD_i) \quad \text{Domestic/import Armington,} \tag{93}$$

$$Q^M_{is} = \text{CES}(Q_{i1s}, \ldots, Q_{iRs}; ESUBM_i) \quad \text{Source Armington,} \tag{94}$$

$$C_{ir} = \text{CDE}(P, Y; SUBPAR, INCPAR) \quad \text{Private consumption,} \tag{95}$$

$$G_{ir} = G(P, Y) \quad \text{Government demand,} \tag{96}$$

$$Q^S_{ir} = Q^D_{ir} \quad \text{Commodity market clearing,} \tag{97}$$

$$F^S_{fr} = \sum_j F^D_{fjr} \quad \text{Factor market clearing,} \tag{98}$$

$$S_r = S(Y_r) \quad \text{Saving,} \tag{99}$$

$$I_r = I(RORE_r) \quad \text{Investment,} \tag{100}$$

$$RORE_r = RORC_r - RORFLEX_r(K^E_r - K^B_r) \quad \text{Expected return,} \tag{101}$$

$$CO2_r = \sum_{i,a} e_{iar} Q_{iar} \quad \text{Emissions,} \tag{102}$$

$$U_r = U(C_r, G_r, S_r) \quad \text{Regional utility,} \tag{103}$$

$$EV_r = E(P^0_r, U^1_r) - E(P^0_r, U^0_r) \quad \text{Welfare.} \tag{104}$$

30 How the Pieces Fit Together

The complete GTAP simulation process can be summarized as

$$\boxed{\text{Database} + \text{Parameters} + \text{Equations} + \text{Closure} + \text{Shock} \quad \Longrightarrow \quad \text{New General Equilibrium.}} \tag{105}$$

More explicitly,

$$\boxed{\underbrace{\text{Benchmark Data}}_{\text{Initial economy}} \rightarrow \underbrace{\boldsymbol{\theta}}_{\text{Behavior}} \rightarrow \underbrace{F(\cdot) = 0}_{\text{CGE equations}} \rightarrow \underbrace{\text{Closure}}_{\text{EXO/ENDO}} \rightarrow \underbrace{\text{Shock/Swap}}_{\text{Experiment}} \rightarrow \underbrace{\text{Solution}}_{\text{Counterfactual equilibrium}}.} \tag{106}$$

31 Key Interpretation for Students

Component	Question Answered
Sets	Who and what exist in the model?
Variables	What can change?
Parameters	How strongly do agents respond?
Equations	How are variables economically connected?
Closure	What is given and what must the model solve?
Shock	What changes from outside the model?
Swap	Which variable becomes the target and which becomes the adjustment mechanism?
Solution	What new equilibrium is consistent with all equations simultaneously?
Welfare	Who gains or loses from the counterfactual?